

Abstract

Measuring Dynamic Trust for Human-AI Complementarity: Toward a Longitudinal Evaluation Framework

Gonzalez et al. (2026) argue that evaluating human-AI teaming requires methods that assess trust calibration and adaptation across interactions, not just single-turn accuracy, yet such methods remain underdeveloped. Complementarity cannot be assumed. Recent evidence shows that human-AI teams frequently perform worse than the stronger individual contributor (Vaccaro, Almaatouq, & Malone, 2024), motivating longitudinal measures that explain when calibrated trust actually produces complementarity. We argue that longitudinal trust calibration provides a behavioral lens for evaluating dynamic alignment between humans and AI.

We present a longitudinal evaluation framework integrating behavioral and subjective trust measures from human-human team science and human-machine trust research into a common approach for evaluating dynamic human-AI teams.

Human-human team science contributes validated behavioral markers of trust, including grounding and repair in communication, transactive memory, and backup behavior, that capture how trust is enacted during collaboration (Clark & Brennan, 1991; Lewis, 2003; Porter et al., 2003). Human-machine trust research complements these measures with prediction of AI error boundaries, reliance and override behavior, and longitudinal trust ratings that capture calibration across repeated interactions (Bansal et al., 2019; McGuirl & Sarter, 2006; Yang, Schemanske, & Searle, 2021).

Building on these complementary traditions, we define dynamic trust as trust continuously updated through interaction and expressed through observable reliance. The framework organizes trust into a four-stage feedback loop consisting of observable behavior, trust calibration, adaptive reliance and task allocation, and updated behavior across successive interactions. It emphasizes behavioral measures from Crew Resource Management marker systems (Salas et al., 2006), while treating convergence and divergence with subjective trust ratings as diagnostically meaningful.

The framework provides a common language for evaluating how calibrated trust contributes to dynamic alignment and human-AI complementarity over time. We seek workshop feedback on which trust mechanisms, and which combinations of behavioral and subjective measures, best transfer to and capture complementarity in human-AI teams.